

Linux Boot Process - Complete Notes

Overview

The Linux boot process consists of 4 main stages that occur from power-on to login:

1. **BIOS** - Hardware initialization and bootloader location
 2. **Bootloader** - Kernel loading with parameters
 3. **Kernel** - Device/memory initialization and init process startup
 4. **Init** - Essential system services management
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Stage 1: BIOS (Basic Input/Output System)

Purpose

- Initializes hardware components
- Performs Power-on Self Test (POST) to verify hardware integrity
- Locates and loads the bootloader

Process

- **Boot Block Search**: BIOS searches for boot block on hard drive
- **Partition Table Check**: Looks at Master Boot Record (MBR) or GPT
- **MBR Location**: First sector of hard drive (first 512 bytes)
- **MBR Contents**: Code to load another program that loads the bootloader

UEFI Alternative

- **Full Form**: Unified Extensible Firmware Interface
- **Purpose**: Modern successor to BIOS
- **Storage**: Uses .efi files stored on EFI system partition
- **Adoption**:
 - Macintosh machines use EFI booting
 - Windows has mostly moved to UEFI
 - GPT format designed for EFI use
- **Compatibility**: GPT disks include "protective MBR" for BIOS compatibility

Stage 2: Bootloader

Main Responsibilities

1. Boot into operating system (Linux or non-Linux)
2. Select kernel to use
3. Specify kernel parameters

Common Bootloaders

- GRUB (most common for Linux)
- LILO
- efilinux
- coreboot
- SYSLINUX

GRUB Boot Parameters

Access via GRUB menu using 'e' key during startup:

- `initrd` - Location of initial RAM disk
- `BOOT_IMAGE` - Kernel image location
- `root` - Root filesystem location
 - Example: `/dev/sda1` or UUID representation
- `ro` - Mounts filesystem in read-only mode
- `quiet` - Suppresses background boot messages
- `splash` - Enables splash screen display

Stage 3: Kernel

Initial Challenge

Chicken-and-egg problem: Kernel manages hardware but doesn't have all drivers available during bootup.

Solution: Temporary Root Filesystem

Historical: initrd (Initial RAM Disk)

- Temporary root filesystem containing essential modules
- Kernel mounts `initrd` → loads bootup drivers → replaces with actual root filesystem

Modern: `initramfs` (Initial RAM Filesystem)

- Built into kernel itself
- Loads necessary drivers for real root filesystem
- No need to locate separate `initrd` file

Root Filesystem Mounting Process

1. **Initial Mount:** Root partition mounted in **read-only mode**
 2. **Integrity Check:** `fsck` runs safely to check system integrity
 3. **Remount:** Root filesystem remounted in **read-write mode**
 4. **Init Launch:** Kernel locates and executes `init` program
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Stage 4: Init

Role

- First process started by kernel
- Starts all other essential system services
- Process ID (PID) = 1

Three Major Init Implementations

1. System V `init` (`sysv`)

- **Type:** Traditional `init` system
- **Method:** Sequential start/stop of processes
- **Control:** Uses startup scripts
- **States:** Machine state defined by runlevels
- **Behavior:** Each runlevel starts/stops machine differently

2. Upstart

- **Usage:** Found on older Ubuntu installations
- **Concept:** Jobs and events based system
- **Method:** Starts jobs that perform actions in response to events
- **Architecture:** Event-driven process management

3. Systemd

- **Status:** New standard for init
 - **Approach:** Goal-oriented system
 - **Method:** Defines goals and satisfies dependencies to complete them
 - **Advantage:** Parallel service startup, faster boot times
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Key Concepts Summary

Boot Sequence Flow

Power On → BIOS/UEFI → Bootloader → Kernel → Init → System Services

Critical Files/Locations

- **MBR:** First 512 bytes of hard drive
- **EFI System Partition:** Contains .efi files for UEFI
- **Kernel Image:** Location specified in BOOT_IMAGE parameter
- **Root Filesystem:** Location specified in root parameter

Modern vs Legacy

- **Legacy:** BIOS + MBR + initrd + SysV init
- **Modern:** UEFI + GPT + initramfs + systemd

Troubleshooting Access Points

- **GRUB Menu:** Press 'e' during startup to view/edit parameters
- **Boot Parameters:** Modify kernel behavior during startup
- **Init Selection:** Different init systems for different distributions